

Local Multi-Project Docker Dev Environment — Clean Setup Guide

Tested working process. Use this for future machines or when adding new projects.

Stack: WSL/Ubuntu host with existing PHP + MySQL → Docker for Traefik (reverse proxy), Nginx, PHP-FPM (per project), and phpMyAdmin. MySQL stays on the host, not containerized.

Part 1 — Install Docker (one-time)

```
bash

# Remove any old/conflicting packages
sudo apt remove docker docker-engine docker.io containerd runc

# Add Docker's official repo
sudo apt update
sudo apt install ca-certificates curl gnupg -y
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
sudo chmod a+r /etc/apt/keyrings/docker.gpg

echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gp
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin doc

# Run docker without sudo
sudo usermod -aG docker $USER
newgrp docker

# Verify
docker --version
docker compose version
docker run hello-world
```

Part 2 — Free up port 80 on the host

If Apache (or any web server) is running on the host, stop it — Traefik will own port 80:

```
bash

sudo service apache2 stop
sudo systemctl disable apache2
```

Existing host PHP/MySQL installs are fine to keep — PHP just won't run the app anymore (the container's PHP is authoritative), and MySQL stays as the shared database.

Part 3 — Configure host MySQL for Docker access

3a. Find the correct config file

The setting we need is `bind-address`, in the **server** config — NOT `/etc/mysql/mysql.conf.d/mysql.cnf` (that's the client config). Find it with:

```
bash

sudo grep -r "bind-address" /etc/mysql/
```

It's typically `/etc/mysql/mysql.conf.d/mysqld.cnf`.

3b. Set bind-address to allow external connections

```
bash

sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
```

Change:

```
bind-address = 127.0.0.1
```

to:

```
bind-address = 0.0.0.0
```

Leave `mysqlx-bind-address` alone — that's for the separate X Protocol port (33060), unrelated to normal MySQL connections.

Restart MySQL:

```
bash
```

```
sudo service mysql restart
```

Confirm it's listening on all interfaces:

```
bash  
sudo ss -tlnp | grep 3306
```

You want to see `0.0.0.0:3306`, not `127.0.0.1:3306`.

3c. Create a MySQL user Docker containers can actually use

Important MySQL concept: an account is the pair `username@host`, not just the username.

`'jubaer'@'localhost'` and `'jubaer'@'%'` are two entirely separate accounts. Docker containers connect from their own container IP (e.g. `172.18.0.3`), not `localhost` — so a `localhost`-only account will always be rejected with "Host '...' is not allowed to connect."

If you already have a `'user'@'localhost'` account and want one account total:

```
bash  
sudo mysql
```

```
sql  
  
RENAME USER 'jubaer'@'localhost' TO 'jubaer'@'%';  
FLUSH PRIVILEGES;  
EXIT;
```

If you'd rather keep `localhost` access AND add Docker access as a separate account:

```
sql  
  
CREATE USER 'jubaer'@'%' IDENTIFIED BY 'yourpassword';  
GRANT ALL PRIVILEGES ON *.* WITH GRANT OPTION;  
FLUSH PRIVILEGES;  
EXIT;
```

Verify:

```
bash  
sudo mysql -e "SELECT user, host FROM mysql.user WHERE user='jubaer';"
```

Part 4 — Create the shared Docker network

```
bash

docker network create proxy
```

Part 5 — Traefik (shared reverse proxy, always on)

Important: use `traefik:v3.7` (or newer) — NOT `v3.0`–`v3.5`. Docker Engine 29+ raised its minimum supported API version, and Traefik versions before 3.6.1 hardcode an old API version in their Docker client, causing a permanent `client version 1.24 is too old` error that breaks all routing. This was fixed in Traefik 3.6.1+.

```
bash

mkdir -p ~/docker/traefik
nano ~/docker/traefik/docker-compose.yml
```

Paste:

```
yaml

services:
  traefik:
    image: traefik:v3.7
    container_name: traefik
    restart: unless-stopped
    command:
      - "--providers.docker=true"
      - "--providers.docker.exposedbydefault=false"
      - "--entrypoints.web.address=:80"
      - "--api.dashboard=true"
      - "--api.insecure=true"    # dashboard at :8080, local dev only
    ports:
      - "80:80"
      - "8080:8080"
    volumes:
      - /var/run/docker.sock:/var/run/docker.sock:ro
    networks:
      - proxy

networks:
  proxy:
    external: true
```

Start it:

```
bash

cd ~/docker/traefik
docker compose up -d
docker logs traefik --tail 20
```

Logs should be clean. Check `docker exec traefik traefik version` shows `3.6.x` or higher.

Dashboard: `http://localhost:8080/dashboard/`

Part 6 — phpMyAdmin (shared, always on)

```
bash

mkdir -p ~/docker/shared-phpmyadmin
nano ~/docker/shared-phpmyadmin/docker-compose.yml
```

Paste (type the backticks directly in the editor — don't generate this file via a bash heredoc with escaped backticks, that corrupts the YAML):

```
yaml

services:
  phpmyadmin:
    image: phpmyadmin:latest
    container_name: phpmyadmin
    restart: unless-stopped
    environment:
      - PMA_HOST=host.docker.internal
      - PMA_PORT=3306
    extra_hosts:
      - "host.docker.internal:host-gateway"
    networks:
      - proxy
    labels:
      - "traefik.enable=true"
      - "traefik.http.routers.phpmyadmin.rule=Host(`phpmyadmin.test`)"
      - "traefik.http.services.phpmyadmin.loadbalancer.server.port=80"

networks:
  proxy:
    external: true
```

Start it:

```
bash

cd ~/docker/shared-phpmyadmin
docker compose up -d
docker ps
```

Part 7 — Add hosts entries (Windows side)

Open Notepad as **Administrator**, edit:

```
C:\Windows\System32\drivers\etc\hosts
```

Add:

```
127.0.0.1  phpmyadmin.test
```

Flush DNS if needed (Windows Command Prompt, not WSL):

```
ipconfig /flushdns
```

Visit <http://phpmyadmin.test>, log in with the [\(%\)](#)-host MySQL user from Part 3c.

Part 8 — New project template

```
bash

mkdir -p ~/projects/myproject/docker/nginx ~/projects/myproject/docker/php
cd ~/projects/myproject
```

Dockerfile:

```
dockerfile
```

```
FROM php:8.3-fpm
```

```
RUN apt-get update && apt-get install -y \  
    git unzip libzip-dev libpng-dev libonig-dev libxml2-dev \  
    && docker-php-ext-install pdo pdo_mysql zip gd mbstring xml \  
    && apt-get clean && rm -rf /var/lib/apt/lists/*
```

```
COPY --from=composer:latest /usr/bin/composer /usr/bin/composer
```

```
COPY docker/php/www.conf /usr/local/etc/php-fpm.d/www.conf
```

```
WORKDIR /var/www/html
```

docker/nginx/default.conf:

```
nginx  
  
server {  
    listen 80;  
    index index.php index.html;  
    root /var/www/html/public;  
    client_max_body_size 20M;  
  
    location / {  
        try_files $uri $uri/ /index.php?$query_string;  
    }  
  
    location ~ /\.php$ {  
        fastcgi_pass app:9000;  
        fastcgi_index index.php;  
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;  
        include fastcgi_params;  
    }  
  
    location ~ /\.ht {  
        deny all;  
    }  
}
```

docker/php/www.conf (caps RAM per project — tune `max_children` per project's expected traffic):

```
ini
```

```
[www]
user = www-data
group = www-data
listen = 9000
listen.owner = www-data
listen.group = www-data

pm = ondemand
pm.max_children = 5
pm.process_idle_timeout = 10s
pm.max_requests = 500
```

docker-compose.yml (replace `myproject` throughout with your actual project name):

```
yaml
```



```

services:
  app:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: myproject_app
    volumes:
      - ./var/www/html
    environment:
      - DB_HOST=host.docker.internal
    extra_hosts:
      - "host.docker.internal:host-gateway"
    expose:
      - "9000"
    networks:
      - default

  nginx:
    image: nginx:stable-alpine
    container_name: myproject_nginx
    restart: unless-stopped
    volumes:
      - ./var/www/html
      - ./docker/nginx/default.conf:/etc/nginx/conf.d/default.conf
    depends_on:
      - app
    networks:
      - proxy
      - default
    labels:
      - "traefik.enable=true"
      - "traefik.http.routers.myproject.rule=Host(`myproject.test`)"
      - "traefik.http.services.myproject.loadbalancer.server.port=80"

networks:
  proxy:
    external: true
  default:
    driver: bridge

```

.env (in your Laravel project root — merge these values in):

```
DB_CONNECTION=mysql
DB_HOST=host.docker.internal
DB_PORT=3306
DB_DATABASE=your_db_name
DB_USERNAME=jubaer
DB_PASSWORD=yourpassword
```

Add to Windows hosts file:

```
127.0.0.1 myproject.test
```

Build and run:

```
bash

docker compose up -d --build
docker compose exec app composer install
docker compose exec app php artisan key:generate
docker compose exec app php artisan migrate
```

Visit <http://myproject.test>

Daily workflow

```
bash

# After reboot, once:
sudo service mysql start
cd ~/docker/traefik && docker compose up -d
cd ~/docker/shared-phpmyadmin && docker compose up -d

# Per project, as you work on it:
cd ~/projects/myproject && docker compose up -d
# when done:
docker compose down
```

Troubleshooting reference (issues hit and fixed during setup)

Symptom	Cause	Fix
YAML <code>found unknown escape character</code> error	Backticks written as <code>```</code> via bash heredoc	Edit the file directly in nano, type plain backticks
Plain-text <code>404 page not found</code> on <code>*.test</code> domain	Traefik up, but no router matched the host	Check <code>docker logs traefik</code> ; usually the Docker API version issue below
Traefik logs: <code>client version 1.24 is too old</code>	Docker Engine 29+ incompatible with Traefik <3.6.1's hardcoded Docker client version	Use <code>traefik:v3.6</code> or newer (fixed via auto version negotiation)
phpMyAdmin: <code>Host 'x.x.x.x' is not allowed to connect</code>	MySQL user exists only as <code>'user'@'localhost'</code> , not <code>'user'@'%'</code>	<code>RENAME USER 'user'@'localhost' TO 'user'@'%' ;</code> or create a second <code>%</code> -host account
<code>docker exec <container> env</code> shows a var, but behavior doesn't change	Container wasn't actually recreated with new config	<code>docker compose down && docker compose up -d</code> (not just <code>--force-recreate</code>)